

Amanda Sewwandi

Computer Science Undergraduate

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Profile

I am a Data Science and AI student passionate about transforming raw data into actionable insights and intelligent solutions. My curiosity drives me to explore the intersection of Machine Learning, Deep Learning, and real-world problem solving. With a strong foundation in Python, statistical modeling, and data visualization, I enjoy tackling complex analytical challenges and thrive in collaborative environments where I can combine technical skills with clear communication to deliver impactful results.

Education

Bachelor of Computing (Hons) in Computer Science

2023 – Present

Colombo

University of Sri Jayewardenepura

Focusing on Data Structures, Algorithms, Database Systems, Software Engineering, and Machine Learning.

G.C.E. Advanced Level (2022)

2022

Anula Vidyalaya, Nugegoda

Combined Mathematics — A • Physics — C • Information & Communication Technology — B

Diploma in English

2022

Aquinas College of Higher Education

Completed up to Level 3 with distinction. Developed strong English language proficiency in speaking, writing, and comprehension.

Projects Showcase

Immune Landscape of Tumors

2025

Bioinformatics & Data Visualization

Interactive dashboard exploring tumor immune composition, mutation burden, and patient survival across 20 cancer types using TCGA pan-cancer data. Built for the CFDE Data Visualization Competition.

Python • Streamlit • Plotly • Pandas • NumPy [View Live](#) | [GitHub](#)

BBC News MLOps Pipeline

2025

MLOps & DevOps

End-to-end MLOps system that trains a news classifier, serves it as a REST API, monitors for data drift, and automatically retrains with zero manual intervention. 98.12% accuracy, 1.5ms inference.

Python • PyTorch • Docker • FastAPI • MLflow [GitHub](#)

Student Depression Predictor

2025

Healthcare & Machine Learning

ML model to predict depression risk in students using a Dockerized MLOps pipeline. Includes data preprocessing, model training, testing, and a Streamlit inference app with automated CI/CD.

Python • Scikit-learn • Docker • Streamlit • Pytest [GitHub](#)

IMDB Sentiment Analysis

2024

NLP & Transformers

Binary sentiment classification on IMDB movie reviews using fine-tuned DistilBERT. Outperformed TF-IDF baseline by 3.8%. Includes interactive Streamlit web app for real-time inference.

Python • PyTorch • HuggingFace • Streamlit • Scikit-learn [GitHub](#)

Traffic Sign Classification

2024

Computer Vision

CNN trained to classify 43 types of German traffic signs with 97.70% test accuracy on 12,630 test images. Deployed as a Streamlit web app with real-time image upload and prediction.

Python • TensorFlow • Keras • OpenCV • Streamlit [GitHub](#)

KLOSET — Multi-Vendor E-Commerce

2024

Full Stack — UI/UX Designer

Full-stack multi-vendor fashion e-commerce platform. Role: UI/UX Designer. Designed responsive interfaces and user flows for product listings, vendor dashboard, cart, and checkout.

React • Tailwind CSS • Node.js • Express • Supabase • GitHub

Research & Publications

Task-Aware Mixed-Precision Quantization of Vision-Language Models for Resource-Constrained Robotic Systems 2025 – Present

Final Year Research — In Progress

Investigating efficient compression of large vision-language models (e.g. CLIP, LLaVA) using task-aware mixed-precision quantization to enable real-time inference on edge devices in robotic systems. Explores hardware-aware NAS, activation-aware quantization, and latency-accuracy trade-offs for deployment on Jetson and Raspberry Pi-class hardware.

Comparative Review of CycleGAN and Diffusion Models for Data Augmentation in Real-Time Plant Disease Detection 2026

First Jayewardenepura Computing Symposium 2026 — Published Research Paper

Systematic review comparing CycleGAN and diffusion-based augmentation strategies for improving real-time plant disease detection under data-scarce conditions. Explores generative model performance, training stability, and downstream classifier accuracy.

[Read Paper](#)

Big Data — Powering the Digital Economy 2025

"Did You Know?" Series, Association of Computing Students, USJ — Published Article

Authored a featured article breaking down the 5 V's of Big Data, real-world industry applications, and the privacy and ethical challenges shaping the field.

[Read Article](#)

Leadership & Community

Campus Coordinator — IEEE University of Sri Jayewardenepura 2025

Appointed Campus Coordinator for IEEE at USJ, leading and coordinating all IEEE-affiliated activities across the university. Managed cross-functional teams, oversaw event planning and execution, and served as the liaison between the student branch and IEEE regional bodies. Drove membership growth, organized technical workshops, and mentored junior members.

IEEE USJ — Event & Competition Organizer 2023 – 2024

Organizing Committee

Served on the organizing committees for flagship IEEE events including IEEE Extreme and Japura Extreme programming competitions. Coordinated logistics, managed participant registration, and supported technical operations.

Sasnaka Sansada — Teaching & Community Service 2022 – Present

Volunteering

Volunteered teaching mathematics to underprivileged students and organizing community events, helping students build confidence in problem-solving and numeracy skills through hands-on mentoring.

Skills

Technical Skills

- **Programming Languages** (Python, SQL, R)
- **Data Analysis & Manipulation** (Pandas, NumPy, Excel, Data Cleaning, Exploratory Data Analysis (EDA))
- **Data Visualization** (Matplotlib, Power BI)
- **Machine Learning** (Supervised Learning, Unsupervised Learning, Model Evaluation, Feature Engineering)
- **Databases** (MySQL, SQLite, SQL Server)
- **Big Data & Cloud** (Microsoft Azure)
- **Statistics & Mathematics** (Probability, Hypothesis Testing, Regression Analysis, Statistical Analysis)
- **Version Control & Collaboration** (Git, GitHub)
- **Development Tools** (Jupyter Notebook, VS Code, Google Colab, Anaconda)

Soft Skills

Team Leadership • Communication • Problem Solving • Collaboration • Project Management • Strategic Thinking • Mentoring • Time Management • Adaptability • Continuous Learning • Analytical Thinking • Critical Thinking

Languages

English	●●●●●
Sinhala	●●●●●
Tamil	●●●○○

References

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